Using Machine Learning to Predict Clinical Outcomes After Shoulder Arthroplasty with a Minimal Feature Set.

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CLINICAL & AGENTIC AI TAKEAWAY

Machine learning models with both full and minimal feature sets can accurately predict clinical outcomes after shoulder arthroplasty, improving decision-making during surgical consultations.

ABSTRACT

--- REPOSITORY & PUBLISHER WEB METADATA --- Affiliations: KenSci, Seattle, WA, USA. Keywords: Clinical research, Machine Learning, Shoulder Arthroplasty, Predictive Outcomes Analytics, Atsa Outcomes, Rtsa Outcomes Extended Abstract from Repository: Background A machine learning analysis was conducted on 5774 shoulder arthroplasty patients to create predictive models for multiple clinical outcome measures after anatomic total shoulder arthroplasty (aTSA) and reverse total shoulder arthroplasty (rTSA). The goal of this study was to compare the accuracy associated with a full-feature set predictive model (ie, full model, comprising 291 parameters) and a minimal-feature set model (ie, abbreviated model, comprising 19 input parameters) to predict clinical outcomes to assess the efficacy of using a minimal feature set of inputs as a shoulder arthroplasty clinical decision-support tool. Methods Clinical data from 2153 primary aTSA patients and 3621 primary rTSA patients were analyzed using the XGBoost machine learning technique to create and test predictive models for multiple outcome measures at different postoperative time points via the full and abbreviated models. Mean absolute errors (MAEs) quantified the difference between actual and predicted outcomes, and each model also predicted whether a patient would experience clinical improvement greater than the patient satisfaction anchor-based thresholds of the minimal clinically important difference and substantial clinical benefit for each outcome measure at 2-3 years after surgery. Results Across all postoperative time points analyzed, the full and abbreviated models had similar MAEs for the American Shoulder and Elbow Surgeons score (±11.7 with full model vs. ±12.0 with abbreviated model), Constant score (±8.9 vs. ±9.8), Global Shoulder Function score (±1.4 vs. ±1.5), visual analog scale pain score (±1.3 vs. ±1.4), active abduction (±20.4° vs. ±21.8°), forward elevation (±17.6° vs. ±19.2°), and external rotation (±12.2° vs. ±12.6°). Marginal improvements in MAEs were observed for each outcome measure prediction when the abbreviated model was supplemented with data on implant size and/or type and measurements of native glenoid anatomy. The full and abbreviated models each effectively risk stratified patients using only preoperative data by accurately identifying patients with improvement greater than the minimal clinically important difference and substantial clinical benefit thresholds. Discussion Our study showed that the full and abbreviated machine learning models achieved similar accuracy in predicting clinical outcomes after aTSA and rTSA at multiple postoperative time points. These promising results demonstrate an efficient utilization of machine learning algorithms to predict clinical outcomes. Our findings using a

minimal feature set of only 19 preoperative inputs suggest that this tool may be easily used during a surgical consultation to improve decision making related to shoulder arthroplasty.

Keywords: Clinical research, Machine Learning, Shoulder Arthroplasty, Predictive Outcomes Analytics, Atsa Outcomes, Rtsa Outcomes

Study Overview & Background

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Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Machine learning models with both full and minimal feature sets can accurately predict clinical outcomes after shoulder arthroplasty, improving decision-making during surgical consultations.

Extended Repository Abstract

Background A machine learning analysis was conducted on 5774 shoulder arthroplasty patients to create predictive models for multiple clinical outcome measures after anatomic total shoulder arthroplasty (aTSA) and reverse total shoulder arthroplasty (rTSA). The goal of this study was to compare the accuracy associated with a full-feature set predictive model (ie, full model, comprising 291 parameters) and a minimal-feature set model (ie, abbreviated model, comprising 19 input parameters) to predict clinical outcomes to assess the efficacy of using a minimal feature set of inputs as a shoulder arthroplasty clinical decision-support tool. Methods Clinical data from 2153 primary aTSA patients and 3621 primary rTSA patients were analyzed using the XGBoost machine learning technique to create and test predictive models for multiple outcome measures at different postoperative time points via the full and abbreviated models. Mean absolute errors (MAEs) quantified the difference between actual and predicted outcomes, and each model also predicted whether a patient would experience clinical improvement greater than the patient satisfaction anchor-based thresholds of the minimal clinically important difference and substantial clinical benefit for each outcome measure at 2-3 years after surgery. Results Across all postoperative time points analyzed, the full and abbreviated models had similar MAEs for the American Shoulder and Elbow Surgeons score (±11.7 with full model vs. ±12.0 with abbreviated model), Constant score (±8.9 vs. ±9.8), Global Shoulder Function score (±1.4 vs. ±1.5), visual analog scale pain score (±1.3 vs. ±1.4), active abduction (±20.4° vs. ±21.8°), forward elevation (±17.6° vs. ±19.2°), and external rotation (±12.2° vs. ±12.6°). Marginal improvements in MAEs were observed for each outcome measure prediction when the abbreviated model was supplemented with data on implant size and/or type and measurements of native glenoid anatomy. The full and abbreviated models each effectively risk stratified patients using only preoperative data by accurately identifying patients with improvement greater than the minimal clinically important difference and substantial clinical benefit thresholds. Discussion Our study showed that the full and abbreviated machine learning models achieved similar accuracy in predicting clinical outcomes after aTSA and rTSA at multiple postoperative time points. These promising results demonstrate an efficient utilization of machine learning algorithms to predict clinical outcomes. Our findings using a minimal feature set of only 19 preoperative inputs suggest that this tool may be easily used during a surgical consultation to improve decision making related to shoulder arthroplasty.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** Vikas Kumar, Christopher P. Roche, Steve Overman, Ryan W. Simovitch, P. Flurin, T. Wright, J. Zuckerman, Howard D. Routman, Ankur Teredesai
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